A Fourier bound for graceful labelings of $K_n \,\square\, P_r$ and $K_n \,\square\, C_r$

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12 October 2026

1 Results

Write the vertices of $K_n \,\square\, P_r$ and $K_n \,\square\, C_r$ as an $n \times r$ array: Column j is the jth copy of K_n, and row i is a path (or a cycle) through the ith vertex of every copy. There are

$$L = \begin{cases} (r-1)n, & \text{for } K_n \,\square\, P_r, \\ rn, & \text{for } K_n \,\square\, C_r, \end{cases} \qquad \text{row edges, and} \qquad m = r\binom{n}{2} + L \quad \text{edges in all.}$$

A graceful labeling assigns distinct labels in $[0, m]$ to the vertices so that the edge lengths $|f(u) - f(v)|$ are 1, 2, …, m in some order. Throughout, $n \geq 2$; for paths $r \geq 2$, and for cycles $r \geq 3$.

Theorem 1. *Let $x_0 = 4.49340\ldots$ be the root of $\tan x = x$ in $(\pi, \frac{3}{2}\pi)$, and let $\gamma_0 = -2\sin x_0/x_0 = 0.434467\ldots$. If*

$$\gamma_0\Big(m + \frac{1}{2}\Big) \;>\; rn - 1 + 2L,$$

then $K_n \,\square\, P_r$ (respectively $K_n \,\square\, C_r$) is not graceful. In particular, the graphs $K_n \,\square\, P_r$ ($r \geq 2$) and $K_n \,\square\, C_r$ ($r \geq 3$) are not graceful when $n \geq 13$.

The Fourier identity behind Theorem 1 is due to Ma and Yi [3], who used it to bound packings of Golomb rulers. The point here is that the columns of a graceful labeling form such a packing, and the numbers that the packing leaves uncovered are exactly the lengths of the row edges. The identity is elementary, and the proofs below are complete without reference to [3].

A second frequency lowers the uniform bound by one.

Theorem 2. *Let*

$$\Gamma = -\frac{5}{24}\Big(2\sin\frac{24}{5} + \frac{3}{10}\sin\frac{48}{5}\Big) = 0.425964\ldots.$$

If

$$\Gamma\Big(m + \frac{1}{2}\Big) \;>\; \frac{13}{10}(rn - 1) + \frac{43}{30}L,$$

then $K_n \,\square\, P_r$ (respectively $K_n \,\square\, C_r$) is not graceful. In particular, the graphs $K_n \,\square\, P_r$ ($r \geq 2$) and $K_n \,\square\, C_r$ ($r \geq 3$) are not graceful when $n \geq 12$.

	largest known	not graceful for $n \geq$		
	graceful n	Theorem 1	Theorem 2	Theorem 5
$K_n \,\square\, P_2$	5	9	10	
$K_n \,\square\, P_3$	6	11	11	
$K_n \,\square\, P_4$	6	11	11	
$K_n \,\square\, P_5$	5	12	11	
$K_n \,\square\, P_6$	5	12	11	
$K_n \,\square\, P_7$	5	12	12	
$K_n \,\square\, P_8$	4	12	12	
$K_n \,\square\, C_3$	4	13	12	8
$K_n \,\square\, C_4$	6	13	12	
$K_n \,\square\, C_5$	5	13	12	11
$K_n \,\square\, C_6$	4	13	12	
$K_n \,\square\, C_7$	4	13	12	12
$K_n \,\square\, C_8$	4	13	12	

Table 1: Results for small r. In the last three columns, the graph is not graceful when n is at least the number shown.

Using the smallest eigenvalue of the whole graph for odd cycles (Section 6) gives a further improvement for $K_n \,\square\, C_3$ and $K_n \,\square\, C_5$. Table 1 lists the results for small r, together with the largest n for which a graceful labeling is known [2]. The cases $K_n \,\square\, P_2$, $K_n \,\square\, P_3$ and $K_n \,\square\, C_3$ were settled completely by Knuth's programs [2]; they are included only for comparison.

For comparison, the counting argument of answer 81 in [1], with the sharpening $\sum_j (x_{1j} + y_{1j}) \geq r(r-1)$, gives 26 for $K_n \,\square\, P_4$, 28 for $K_n \,\square\, P_5$, and 35 for all $r \geq 2$ (paths) and $r \geq 3$ (cycles).

2 The columns form a packing of Golomb rulers

A set $A = \{a_1 < \cdots < a_n\}$ of integers is a *Golomb ruler* if its $\binom{n}{2}$ differences $a_k - a_l$ $(k > l)$ are distinct.

Consider a graceful labeling f of $K_n \,\square\, P_r$ or $K_n \,\square\, C_r$, and let A_j be the set of labels in column j. Column j is a complete graph, so its edge lengths are exactly the positive differences of A_j. All edge lengths are distinct, and therefore:

1. each A_j is a Golomb ruler;
2. the differences of different columns are distinct;
3. all $r\binom{n}{2} = m - L$ column differences lie in $[1, m]$;
4. the set $\mathcal{R}$ of numbers in $[1, m]$ that are *not* column differences has L elements, and it is exactly the set of lengths of the row edges.

For example, the labeling of $K_5 \,\square\, P_4$ in [2] has columns $\{4, 23, 30, 40, 53\}$, $\{0, 2, 37, 52, 55\}$, $\{7, 8, 16, 50, 54\}$, $\{6, 20, 26, 31, 47\}$. Here $m = 55$, and the 40 column differences are $[1, 55]$ minus

$$\mathcal{R} = \{12, 22, 24, 28, 29, 31, 32, 33, 39, 40, 44, 45, 48, 51, 54\},$$

which are the lengths of the 15 row edges.

3 The identity

For each column, let

$$P_j(\theta) = \sum_{a \in A_j} e^{ia\theta}.$$

The diagonal terms of $|P_j(\theta)|^2 = \sum_{a,a' \in A_j} e^{i(a-a')\theta}$ contribute n, and each unordered pair $\{a, a'\}$ contributes $2\cos\big((a-a')\theta\big)$. Summing over the columns and using properties 3 and 4, we obtain

$$\sum_{j=1}^{r} |P_j(\theta)|^2 = rn + 2\sum_{d=1}^{m} \cos(d\theta) - 2\sum_{d \in \mathcal{R}} \cos(d\theta).$$

This is the identity that Ma and Yi [3] use for packings of Golomb rulers; here property 4 identifies the uncovered numbers with the row lengths. The middle sum is expressed by the Dirichlet kernel

$$D_m(\theta) = 1 + 2\sum_{d=1}^{m} \cos(d\theta) = \frac{\sin\big((m+\tfrac{1}{2})\theta\big)}{\sin(\theta/2)}.$$

Lemma 3. *For every graceful labeling and every real θ,*

$$\sum_{j=1}^{r} |P_j(\theta)|^2 = rn - 1 + D_m(\theta) - 2\sum_{d \in \mathcal{R}} \cos(d\theta).$$

Consequently $rn - 1 + D_m(\theta) + 2L \geq 0$.

Proof. The identity is the display preceding the definition of D_m. Its left side is nonnegative, and each of the L terms $-\cos(d\theta)$ with $d \in \mathcal{R}$ is at most 1. □

For the labeling of $K_5 \,\square\, P_4$ in Section 2 and $\theta = x_0/(m+\frac{1}{2})$, the four values $|P_j(\theta)|^2$ are approximately 3.098, 2.428, 1.066, 6.902, with sum 13.494; on the right side, $rn - 1 = 19$, $D_{55}(\theta) = -24.120$ and $\sum_{d \in \mathcal{R}} \cos(d\theta) = -9.307$.

4 One frequency

The only free parameter in Lemma 3 is θ, and we want $D_m(\theta)$ to be as negative as possible. If $\theta = x/(m+\frac{1}{2})$ for a fixed x, then $\sum_{d=1}^{m} \cos(d\theta)$ is approximately $(m/x)\int_0^x \cos t\, dt = m \sin x/x$. The function $\sin x/x$ attains its smallest value $-0.2172\ldots$ at $x = x_0$. So the numbers $1, \ldots, m$, each used once, make D_m about $-0.434\, m$; the row lengths can compensate at most $2L$, and $2L$ grows only linearly in n while m grows quadratically.

Lemma 4. *If $\sin x < 0$, $0 < x < (2m+1)\pi$ and $\theta = x/(m+\frac{1}{2})$, then*

$$D_m(\theta) \;\leq\; \frac{2\sin x}{x}\Big(m + \frac{1}{2}\Big).$$

Proof. We have $D_m(\theta) = \sin x/\sin y$, where $y = x/(2m+1)$ lies in $(0, \pi)$. Since $0 < \sin y \leq y$ and $\sin x < 0$, we get $\sin x/\sin y \leq \sin x/y = (2m+1)\sin x/x$. □

For $\pi < x < 2\pi$ the factor $-2\sin x/x$ is largest at $x = x_0$, where it equals γ_0. One may avoid the irrational number x_0: The choice $x = 9/2$ gives $-4\sin(9/2)/9 = 0.434457\ldots$, and replacing γ_0 by this value gives the same integer thresholds below.

Proof of Theorem 1. Take $\theta = x_0/(m+\frac{1}{2})$. By Lemma 4, $rn-1+D_m(\theta)+2L \le rn-1+2L-\gamma_0(m+\frac{1}{2})$, which is negative under the hypothesis of the theorem; this contradicts Lemma 3.

To get the uniform statement, substitute the values of m and L. For $K_n \,\square\, P_r$ we have $\gamma_0 m = \gamma_0 rn\frac{n+1}{2} - \gamma_0 n$ and $rn-1+2L = 3rn-2n-1$, so

$$\gamma_0\Big(m+\frac{1}{2}\Big) - (rn-1+2L) = rn\Big(\gamma_0\,\frac{n+1}{2} - 3\Big) + (2-\gamma_0)\,n + \frac{\gamma_0}{2} + 1.$$

For $K_n \,\square\, C_r$ we have $\gamma_0 m = \gamma_0 rn\frac{n+1}{2}$ and $rn-1+2L = 3rn-1$, so

$$\gamma_0\Big(m+\frac{1}{2}\Big) - (rn-1+2L) = rn\Big(\gamma_0\,\frac{n+1}{2} - 3\Big) + \frac{\gamma_0}{2} + 1.$$

When $n \ge 13$ we have $\gamma_0\frac{n+1}{2} \ge 7\gamma_0 = 3.041\ldots > 3$, so both expressions are positive for every $r \ge 1$. □

When $n = 12$ the coefficient $\frac{13}{2}\gamma_0 - 3 = -0.175\ldots$ is negative, so Theorem 1 by itself gives no uniform bound below 13. For $K_n \,\square\, P_4$ the hypothesis reads $110.14 > 109$ when $n = 11$ ($m = 253$, $L = 33$) and $130.56 > 119$ when $n = 12$, but fails, as $91.46 < 99$, when $n = 10$.

5 Two frequencies

Lemma 3 bounds the row term separately for each θ, as if the row lengths could be chosen anew for each frequency. They cannot: $\mathcal{R}$ is one fixed set of L numbers. Adding the identities for two frequencies takes advantage of this, because $-\cos t$ and $-\cos 2t$ are never close to 1 at the same time.

Proof of Theorem 2. Let $x = 24/5$, $a = 3/10$ and $\theta = x/(m+\frac{1}{2})$. Lemma 3 with θ and with 2θ, the second multiplied by a, gives

$$0 \;\le\; (1+a)(rn-1) + D_m(\theta) + a\,D_m(2\theta) + 2\sum_{d\in\mathcal{R}} h(d\theta), \qquad h(t) = -\cos t - a\cos 2t.$$

Writing $c = \cos t$, we have $h(t) = a - c - 2ac^2 \le a + \frac{1}{8a} = \frac{43}{60}$, the maximum of this quadratic in c. So $\sum_{d\in\mathcal{R}} h(d\theta) \le \frac{43}{60}L$, and $2\sum_{d\in\mathcal{R}} h(d\theta) \le \frac{43}{30}L$.

Both $\sin x = -0.99616\ldots$ and $\sin 2x = \sin\frac{48}{5} = -0.17432\ldots$ are negative, and $2x < (2m+1)\pi$ because $m \ge 2$. Lemma 4, applied with x and with $2x$, gives

$$D_m(\theta) + a\,D_m(2\theta) \;\le\; \Big(\frac{2\sin x}{x} + a\,\frac{2\sin 2x}{2x}\Big)\Big(m+\frac{1}{2}\Big) = -\Gamma\Big(m+\frac{1}{2}\Big).$$

Therefore a graceful labeling satisfies $\Gamma(m+\frac{1}{2}) \le \frac{13}{10}(rn-1) + \frac{43}{30}L$, which is the first statement.

For the uniform statement, note that $\frac{13}{10} + \frac{43}{30} = \frac{41}{15}$. For $K_n \,\square\, C_r$,

$$\Gamma\Big(m+\frac{1}{2}\Big) - \frac{13}{10}(rn-1) - \frac{43}{30}L = rn\Big(\Gamma\,\frac{n+1}{2} - \frac{41}{15}\Big) + \frac{\Gamma}{2} + \frac{13}{10},$$

and for $K_n \,\square\, P_r$ the same expression equals

$$rn\Big(\Gamma\,\frac{n+1}{2} - \frac{41}{15}\Big) + \Big(\frac{43}{30} - \Gamma\Big)n + \frac{\Gamma}{2} + \frac{13}{10}.$$

When $n \ge 12$ we have $\Gamma\frac{n+1}{2} \ge \frac{13}{2}\Gamma = 2.7687\ldots > \frac{41}{15} = 2.7333\ldots$, so both expressions are positive for every $r \ge 1$. □

For $K_n \,\square\, P_5$ the hypothesis of Theorem 2 reads $136.10 > 133.27$ when $n = 11$ ($m = 319$, $L = 44$), but fails, as $113.09 < 121.03$, when $n = 10$. Theorem 2 is weaker than Theorem 1 for $K_n \,\square\, P_2$, where L is small compared with rn.

More frequencies help a little more. A numerical search with five or six frequencies suggests that 11 suffices also for $K_n \,\square\, P_7$ and $K_n \,\square\, P_8$; we do not claim this.

6 Odd cycles

Another way to treat the row edges is to put the whole graph into one quadratic form. Let A be the adjacency matrix of the graph and $z_v = e^{if(v)\theta}$. Since the edge lengths are $1, \ldots, m$,

$$z^*Az = 2\sum_{uv\in E} \cos\big((f(u) - f(v))\theta\big) = 2\sum_{d=1}^{m} \cos(d\theta) = D_m(\theta) - 1.$$

On the other hand $z^*Az \ge \lambda_{\min}(A)\, z^*z = \lambda_{\min}(A)\, rn$. The eigenvalues of $K_n \,\square\, C_r$ are the sums of an eigenvalue of K_n ($n-1$ or -1) and an eigenvalue of C_r ($2\cos(2\pi k/r)$); for odd r the smallest is $\lambda_{\min} = -1 - 2\cos(\pi/r)$. Together with Lemma 4, applied with $x = x_0$, this proves:

Theorem 5. *Let r be odd. If $\gamma_0(m + \frac{1}{2}) > \big(1 + 2\cos(\pi/r)\big)rn - 1$, then $K_n \,\square\, C_r$ is not graceful.*

For $r = 5$ and $n = 11$ ($m = 330$) the two sides are 143.59 and 142.99. For even r we have $\lambda_{\min} = -3$, and the same spectral argument gives exactly Theorem 1. For paths $\lambda_{\min} = -1 - 2\cos\frac{\pi}{r+1}$; the two bounds coincide when $r = 2$, and the exact count $L = (r-1)n$ in Theorem 1 is better when $r \ge 3$.

7 Remarks

The limit of the method. The proofs of Theorems 1 and 2 use only properties 1–4 of the columns; the row edges enter only through the number L. In the language of difference triangle sets, properties 1–3 say that the columns, each translated to start at 0, form an $(r, n-1)$ difference triangle set of scope at most m. Whenever such a set exists, no argument based only on the column differences can exclude the corresponding n. For example, Shehadeh, Kingsford and Kschischang [4] give the $(4, 7)$ difference triangle set

0	5	43	49	82	119	123	134
0	17	25	65	96	120	122	132
0	20	23	50	84	112	113	131
0	9	22	54	68	75	110	126

whose 112 differences are distinct and at most 134, while $K_8 \,\square\, P_4$ has $m = 136$. So $K_8 \,\square\, P_4$ cannot be settled by an argument of this kind; it needs more information, such as the structure of the rows.

Parity. When n is odd, every vertex of $K_n \,\square\, C_r$ has even degree $n+1$, and the graph is not graceful if $m \equiv 1$ or 2 (mod 4) [2]. Within the gaps left by Table 1 this excludes $K_9 \,\square\, C_5$, $K_9 \,\square\, C_6$ and $K_{11} \,\square\, C_7$, besides $K_5 \,\square\, C_3$, $K_5 \,\square\, C_6$ and $K_5 \,\square\, C_7$, which are already settled in [2].

References

[1] Donald E. Knuth, *The Art of Computer Programming*, Volume 4, Fascicle 7, errata and addenda, file `all4f7.ps` (12 October 2026).

[2] Donald E. Knuth, "Coups de grâceful graphs," unpublished note (2020; revised 2 January 2026).

[3] Chaohang Ma and Xiangjie Yi, "An almost-covering threshold for Golomb-ruler difference packings," arXiv:2608.13739 (2026).

[4] Mohannad Shehadeh, William Kingsford and Frank R. Kschischang, "New difference triangle sets by an FPGA-based search technique," arXiv:2502.19517 (2025).